

The E3 ubiquitin ligase midline 1 reduces the intrinsic antibacterial activity of hepatocytes by promoting HRG degradation during sepsis

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ARTICLE INFO

Keywords:

Hepatocyte
Mid1
HRG
Antibacterial capacity
Sepsis
Ubiquitination

ABSTRACT

Aims: The ubiquitination system in hepatocytes and its direct role in antibacterial defense during sepsis remain poorly understood. This study investigated the mechanism by which the E3 ubiquitin ligase Midline 1 (Mid1) regulates the degradation of histidine-rich glycoprotein (HRG) during bacterial infection, and a proof-of-concept approach was proposed to block Mid1 and relieve hepatic sepsis.

Material and methods: Transcriptomic analysis of livers from a mouse cecal ligation and puncture (CLP) model was used to reveal changes in the global ubiquitination and Mid1 level. Immunoprecipitation, coupled with mass spectrometry, and native PAGE, was used to assess the proteins that bind to Mid1. The therapeutic effects of Mid1 in vivo were evaluated through hepatocyte-specific siRNAs and inhibitors.

Key findings: Mid1 was significantly upregulated in hepatocytes from both CLP mice and patients with acute infection, accompanied by increased global ubiquitination. Inhibiting Mid1 in hepatocytes enhanced their antibacterial capacity and reduced infection-induced cellular damage *in cellulo* and *in vivo* experiments. Mid1 overexpression increased bacterial survival in AML12 hepatocyte cell lines. Mechanistically, Mid1 physically directly binds the HRG and targets it for ubiquitin-proteasomal degradation. Crucially, HRG, identified as a novel Mid1 substrate, is critical for antibacterial defense—blocking HRG abolished the protective effects of Mid1 knockdown. Hepatocyte-specific Mid1 siRNA and DPP4 inhibitors improved systemic outcomes and survival in septic mice.

Significance: This study highlights that the Mid1-HRG ubiquitination axis is a critical regulator of sepsis progression, suggesting a novel therapeutic strategy to augment the liver's intrinsic antibacterial activity.

Abbreviations

AAV adeno-associated virus
AST aspartate aminotransferase
CHX cycloheximide
CLP cecal ligation and puncture
CR creatinine
CRP C-reactive protein
DPP4 dipeptidyl peptidase-4
HRG histidine-rich glycoprotein
ICAM-1 intercellular cell adhesion molecule 1

IL-1 β interleukin-1 β
IP immunoprecipitation
IRF3 interferon regulatory factor 3
LDH lactate dehydrogenase
LPS lipopolysaccharide
Mid1 midline-1
MS mass spectrometry
PI propidium iodide
PLT platelet
PP2Ac protein phosphatase 2A catalytic subunit
SAA serum amyloid A

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STAT1 signal transducer and activator of transcription 1
 TNF- α tumor necrosis factor alpha

1. Introduction

Sepsis is now defined as a dysregulated host response to infection, resulting in life-threatening organ dysfunction. The disease is critical, and the incidence and mortality rates remain high [1]. Excessive inflammatory responses and cytokine storms during sepsis have been consistently cited as the main causes of high mortality. However, drugs targeting important inflammatory mediators (e.g., TNF- α , IL-1 β , and toll-like receptors) have not yielded satisfactory clinical results in improving the survival rates of sepsis patients [2]. Recent studies have shown that innate immune dysfunction and acquired immunosuppression jointly contribute to multiorgan injury and sepsis-related mortality; thus, a strategy for enhancing the immune response has been proposed. However, relevant studies lack evidence from large-scale randomized controlled trials (RCTs) and are therefore not included in current sepsis treatment guidelines [3]. The treatments recommended by guidelines mainly include resuscitation, organ support, early administration of antibiotics, and control of the infection focus. Nevertheless, specific treatment methods are still lacking.

Histidine-rich glycoprotein (HRG), a hepatocyte-secreted serum protein, circulates at relatively high concentrations in human serum (100–150 $\mu\text{g/ml}$) and has been hailed as the “Swiss Army Knife” of serum proteins owing to its multifunctional properties [4]. During sepsis, the serum HRG concentration decreases [5] and is a reliable biomarker for determining the presence of bacterial infection and the severity of the disease in patients [6]. In the context of infection and inflammation, HRG has diverse protective effects [7], including promoting macrophage phagocytosis [8], restraining the excessive activation of neutrophils [9], inhibiting the formation of insoluble immune complexes (IIC) [10], and neutralizing endotoxins [11]. HRG exerts direct antibacterial effects by disrupting bacterial cell wall integrity in gram-negative bacteria (*Escherichia coli* and *Pseudomonas aeruginosa*) and gram-positive bacteria (*Enterococcus faecalis* and *Staphylococcus aureus*) [12]. Exogenous HRG supplementation (20 mg/kg) significantly improves survival rates in septic mice [9], highlighting its therapeutic potential. Despite these promising findings, large-scale production of medical-grade HRGs remains economically unfeasible, preventing the clinical translation of “exogenous HRG supplementation therapy” to date. The regulatory mechanisms of HRG in hepatocytes during sepsis require further investigation.

Mid1, a RING finger family E3 ubiquitin ligase, plays a pivotal role in diverse disease processes. It is involved in multiple contexts, including asthma (in bronchial epithelial cells) [13], esophagitis (in the esophageal epithelium) [14], and sepsis-induced lung injury (in postcapillary venous endothelial cells) [15]. In spinal cord injury, the inhibition of MID1 transcription promotes macrophage M2 polarization, which in turn favors the reconstruction of the neural stem cell microenvironment [16]. In multiple sclerosis models, Mid1 promotes effector T cell migration to the central nervous system by upregulating the mTOR/microtubule pathway [17]. Phosphatase 2Ac (PP2Ac) and interferon regulatory factor 3 (IRF3) serve as substrates for Mid1-mediated ubiquitination. Ubiquitin-mediated degradation of IRF3 suppresses type I interferon (IFN-I) production and impairs antiviral responses [18]. However, little is known regarding the role of hepatocyte Mid1 in septic liver injury.

In this study, we identified a significant increase in total ubiquitination levels and Mid1 expression in septic liver tissues. Hepatocyte-specific Mid1 knockdown attenuated hepatocyte damage and enhanced bacterial clearance both *in cellulo* and *in vivo*. We now demonstrated for the first time that HRG is a novel substrate of Mid1-mediated ubiquitination. Therapeutically, hepatocyte-targeted Mid1 siRNA and an oral DPP4 inhibitor effectively suppressed hepatic Mid1 expression. This intervention elevated serum HRG levels, improved liver

function, and enhanced survival in septic mice. In summary, these findings reveal that the Mid1-HRG axis is as a novel regulator of liver sepsis progression, suggesting a crucial target for sepsis therapies.

2. Material and methods

2.1. Bacterial strains and culture conditions

The *E. coli* and MRSA (ATCC 43300) strains were cultured overnight in Luria-Bertani (LB) medium at 37 °C with shaking at 200 rpm. The following morning, they reached the logarithmic growth phase. Bacterial growth was monitored by measuring the optical density at 600 nm (OD600), where an OD600 of 1.0 corresponded to approximately 1×10^9 colony-forming units (CFU)/ml for *E. coli* [19] and an OD600 of 0.5 to approximately 1.5×10^8 CFU/ml for MRSA [20]. The bacterial suspensions were then diluted to the required concentrations for subsequent *in vivo* (animal model) and *in cellulo* (cell model) experiments.

2.2. Animals

Male C57BL/6 J mice (6–8 weeks old) were obtained from Charles River Laboratory (Beijing, China) and housed in a specific pathogen-free (SPF) facility at the Animal Experiment Center of the Army Medical Center (Chongqing, China). The mice were maintained under controlled conditions (22–25 °C, 12-h light/dark cycle) with adequate ventilation and free access to food and water. All animal experiments were approved by the Animal Ethics Committee of the Army Medical University (approval no. AMUWEC20247132) and conducted in compliance with the guidelines of the Institutional Animal Care and Use Committee.

2.3. Human liver tissue samples

Healthy human liver tissue samples were obtained from macroscopically normal residual tissue of patients undergoing resection for benign liver diseases. Infected liver tissue samples were obtained from patients diagnosed with acute cholangitis, a condition resulting from the combination of bacterial infection and biliary obstruction [21]. This study was approved by the Ethics Committee of the Army Medical Center (approval no. YYLS2025038). Written informed consent was obtained from all participants in accordance with the principles outlined in the Declaration of Helsinki. The donors consented to the pseudonymized use of their tissue samples and clinical data (sex, age, diagnosis) for research and publication purposes. No compensation was provided for participation.

2.4. Mouse sepsis and peritonitis models

Cecal ligation and puncture (CLP) model: Following laparotomy, the cecum was exposed, and the fecal contents were concentrated in the distal cecum. The cecum was ligated and punctured approximately 1.2 cm from the distal end, allowing a small amount of fecal material to extrude before being returned to the abdominal cavity. The incision was then sutured.

LPS-induced peritonitis model: Lipopolysaccharide (LPS) was dissolved in sterile saline and administered intraperitoneally (10 mg/kg). After 12 or 24 h, the mice were euthanized, and specimens were collected for further analysis.

Mouse serum was collected and analyzed via an automated biochemical analyzer (Beckman Coulter) to measure the following parameters. Liver function markers included alanine aminotransferase (ALT), aspartate aminotransferase (AST), total bilirubin (TB), lactate dehydrogenase (LDH), and gamma-glutamyl transferase (GGT). Kidney function marker: Creatinine (Crea). Hematological parameter: Platelet count (PLT). HRG levels were quantified by immunoblotting and ELISA (RayBiotech, Cat#Q9ESB3).

2.5. Histopathological analysis

Mouse liver samples were fixed in 4 % paraformaldehyde, dehydrated, embedded in paraffin, and sectioned. For morphological assessment, hematoxylin and eosin (H&E) staining was performed.

Immunohistochemistry was performed using a two-step universal kit (ZSGB-BIO, Cat#PV-8000). Briefly, the sections were dewaxed, rehydrated, and treated with PBST (PBS containing 0.1 % Triton X-100) for 10 min, followed by antigen retrieval. Endogenous peroxidase activity was blocked by incubating the sections with a peroxidase inhibitor at room temperature (RT) for 10 min. An anti-Mid1 (Abclonal, Cat#A7291) antibody was applied, and the samples were incubated overnight at 4 °C. After washing with PBS, an enzyme-linked sheep anti-mouse/rabbit IgG polymer was added, and the mixture was incubated at 37 °C for 20 min. After another PBS wash, the sections were developed with DAB chromogen, counterstained with hematoxylin, and differentiated until the nuclei appeared blue. Finally, the sections were dehydrated, cleared, and mounted for microscopic analysis.

For immunofluorescence staining, endogenous peroxidases were blocked as described for IHC. The following primary antibodies were used: anti-Mid1 rabbit antibody (Abclonal, Cat#A7291), anti-CD3 rabbit antibody (Abcam, Cat#ab16669), anti-CD15 rabbit antibody (Abcam, ab15377), anti-CD68 rabbit antibody (Abcam, Cat#ab125212), and anti-HRG mouse antibody (Santa Cruz, Cat#sc-398,239). The sections were incubated overnight at 4 °C with primary antibodies, washed with PBS, and then incubated with the following secondary antibodies: Alexa Fluor 488-labelled anti-mouse IgG (Thermo Fisher, Cat#A-21202) or Alexa Fluor 647-labelled anti-rabbit IgG (Thermo Fisher, Cat#A-21245). After 2 h at RT, the nuclei were stained with DAPI (5 min, RT). The sections were washed, mounted with an anti-fade medium, and coverslipped. The fluorescence signals were detected via an OLYMPUS superresolution microscope system.

2.6. Isolation of primary hepatocytes

Primary hepatocytes were isolated using an established protocol [22]. Briefly, the liver tissue was digested to create a hepatocyte suspension, which was then filtered through a 70 µm cell strainer. The filtrate was centrifuged at 50 ×g for 2 min at 4 °C to pellet the hepatocytes. The resulting cell pellet was resuspended in the appropriate medium for subsequent experiments.

2.7. Cell culture and transfection

The following cell lines were cultured under standard conditions (37 °C, 5 %CO₂, humidified atmosphere): AML12 (Alpha mouse liver 12; Procell Life Science Technology Co., Ltd., Wuhan, China) in AML12-specific medium, Kupffer macrophages in RPMI-1640 medium, and liver sinusoidal endothelial cells (LSECs) and hepatic stellate cells (JS1) in DMEM-H medium. All culture media were supplemented with 10 % fetal bovine serum and 100 U/ml penicillin-streptomycin.

AML12 cells were transfected with lentiviral vectors for delivery via gene-editing systems. Murine Mid1 knockdown: Cells were transduced with a siRNA-containing lentivirus (target sequence: GACCGATGAC-CAGTTAATCTGTGTTTTT) constructed by Beijing Qingke Biotechnology Co., Ltd. Murine Mid1 overexpression: *Mid1* transcript variant 3 sequence from NCBI GenBank (NM_001290504.1) was cloned and inserted into lentiviral vector particles (GenePharma Technology Co., Ltd.). The cells were transduced with a Mid1-overexpressing lentivirus. The optimal multiplicity of infection (MOI) was determined through preliminary screening. Stable cell lines were established according to the manufacturers' protocols and subsequently validated.

2.8. In cellulo model

An *in cellulo* coculture system was established using AML12

hepatocytes, Kupffer macrophages, liver sinusoidal endothelial cells (LSECs), and hepatic stellate cells (JS1). The cells were treated with or without *E. coli* (MOI of 10:1), LPS (1 µg/ml; Sigma, Cat#L5293), TAK-243 (1 µM; MCE, Cat#HY-100487), CHX (10 µg/ml; MCE, Cat#HY-12320), or MG132 (10 µM; MCE, Cat#HY-13259). Notably, antibiotics were excluded from the medium during the bacterial challenge experiments. After treatment, the cells were harvested for RT-qPCR and Immunoblot analysis.

2.9. Heparin-bacterial competition assay for HRG binding

The method employed for the heparin-bacterial competition assay to evaluate HRG binding was adapted from the literature [23]. In summary, AML12 cells (2 × 10⁶) were cultured in antibiotic-free medium for 12 h and subsequently infected with *E. coli* (1.5 × 10⁸ CFU) per well. Heparin (0, 10, 100, 200 and 400 µg/ml) was then added to each well and incubated at 37 °C for 2 h. Following incubation, the cells were centrifuged at 300 ×g for 5 min, and the supernatants were collected. These supernatants were utilized directly for soluble protein analysis, while the cell pellets were lysed in RIPA buffer. Equal volumes of each supernatant along with equivalent amounts of proteins from the pellet samples were subjected to immunoblotting analysis.

2.10. Immunoprecipitation

Cultured cells were lysed on ice for 10 min in RIPA buffer supplemented with protease inhibitors, after which the lysate was centrifuged. The total protein in the supernatant was collected, quantified and incubated with an anti-Mid1 antibody (Thermo Scientific, Cat# PA5-36305 and Abclonal, Cat# A7291), an anti-ubiquitin antibody (Santa Cruz, Cat# sc-8017), and the corresponding isotype IgG (CST, Cat# 3900). A Pierce Coimmunoprecipitation Kit (Thermo Scientific, Cat# 88828) was used according to the manufacturer's protocol, followed by immunoblot analysis.

2.11. Native PAGE protocol

The cell samples were collected and lysed with a non-denaturing lysis buffer (Sangon Biotech, Cat# C51001). The lysate was centrifuged to pellet the cellular debris, and the supernatant was mixed with native PAGE protein loading buffer (Beyotime, Cat# P0016) to prepare the samples. An 8 % SDS-free acrylamide gel was used for separation. Electrophoresis was performed at a low temperature using Tris-Glycine electrophoresis buffer (pH 8.8), followed by immunoblot analysis. In parallel, the gel was stained with Coomassie Brilliant Blue (Beyotime, Cat# P0003) to visualize the protein bands.

2.12. Immunoblot analysis

Tissue samples and cultured cells (particularly diluted mouse serum and cultured supernatant for HRG analysis) were lysed via RIPA buffer (CST, #9806) supplemented with protease and phosphatase inhibitors (CST, Cat#5872). The lysates were centrifuged for clarification, and the supernatants were collected for immunoblotting. Equal amounts of total protein were loaded onto Tris-glycine extended stain-free polyacrylamide gels. Electrophoresis was performed to separate the proteins under denaturing conditions. Total protein, measured using a ChemiDoc Touch Imaging System (Bio-Rad), serve as a stain-free loading control for normalization. After the proteins were transferred to a PVDF or nitrocellulose membrane, the membrane was blocked to prevent nonspecific binding of antibodies binding prior to immunoblotting. The PVDF membranes were incubated with the following primary antibodies: (HRG (Abclonal, Cat# A8431), Mid1 (Thermo Scientific, Cat# PA5-36305), Capza1 (Abclonal, Cat# A3776), PP2Acα (Santa Cruz, Cat# sc-80,665), total ubiquitination (PTMBIO, Cat# PTM-5798), ubiquitin (Abclonal, Cat# A19686), GAPDH (CST, Cat# 5174), β-actin

(CST, Cat# 4967), Albumin (ABclonal, Cat# A24161), and HNF4 α (Biotime, Cat# AF7122). Then, secondary antibodies and chemiluminescent detection protocols were used as previously described. Immunoreactive protein bands were visualized via the ChemiDoc Touch System (Bio-Rad, USA).

2.13. RT-qPCR

Total RNA was extracted from the samples using TRIzol reagent, following the manufacturer's protocol. Complementary DNA (cDNA) was synthesized via the HiScript II One-Step RT-PCR Kit (Vazyme, Cat# RR014BP611). Gene expression was analyzed via quantitative PCR (qPCR) with ChamQ SYBR qPCR Master Mix (Vazyme, Cat# Q321) on a CFX96 Touch Real-Time PCR System (Bio-Rad). GAPDH served as the internal control, and relative gene expression levels were calculated via the $2^{-\Delta\Delta Ct}$ method. The mRNA expression of the target gene was compared to that of the control group, following previously established protocols.

The sequences of primers used were as follows:

mMid1F: CCAGGTCCTAATCCCGAAATCA;
mMid1R: GGGTTGGGAGCTGTAGGTAATC.
mGAPDH F: CAGTGGCAAAGTGGAGATTGTTG;
mGAPDH R: TCGCTCCTGGAAGATGGTGAT.
hMid1F: CTGACCTGCCCTATTGTCTG;
hMid1R: GCACAGTGTGATACTAGGATGC.

2.14. Flow cytometry

A PE Annexin V Apoptosis Detection Kit with 7-AAD (BioLegend, Cat# 640934) was used to identify apoptotic and necrotic cells. After 12 h of coculture with *E. coli*, the cells were collected and stained according to the manufacturer's protocol, followed by incubation in the dark at room temperature. The stained cells were then analyzed via an ACEA NovoCyte™ flow cytometer, and the data were processed with FlowJo V10.3 software.

2.15. Viral vector construction, siRNA preparation and administration

An adenoviral vector was constructed via the GV698 promoter. The sequence for *Mid1* was: GACCGATGACCAAGTAACTGTGTTT. For in vivo delivery, 100 μ l of adeno-associated virus (AAV; 2×10^{12} viral particles/ml) was administered intravenously, with AAV-NC serving as the negative control. The infection model was successfully established three weeks after injection.

The following Cy3-labelled siRNAs were synthesized by GenePharma Technology Co. (Shanghai, China): *Mid1*-targeting siRNA (mid1-siRNA; antisense: 5'-AUAAGAUUGGUGAGAUUACUC-3'); scrambled control siRNA (siN. C; antisense: 5'-ACGUGACACGUUCGGAGAATT-3'). siRNAs were administered intravenously (60 μ g per 20 g body weight) 12 h prior to CLP induction.

2.16. Bacterial viability analysis

To assess bacterial viability, the cells were cultured in media without antibiotics and infected with MRSA or *E. coli* at a multiplicity of infection (MOI) of 2 for 18 h. Bacterial viability was then evaluated via the LIVE/DEAD Bacterial Viability Kit (Invitrogen, Cat# L7012), following the manufacturer's protocol (live bacteria: green fluorescence; dead bacteria: red fluorescence). The fluorescence signals were visualized and quantified via an Olympus superresolution microscope system.

2.17. Transcriptomic analyses

Total RNA was extracted from the livers of CLP ($n = 5$), Sham ($n = 3$), CLP_NC ($n = 3$), and CLP_shMid1 ($n = 3$) mice for CLP vs Sham and CLP_NC vs CLP_shMid1 analysis. RNA sequencing was performed on the

Illumina NovaSeq 6000 platform (Illumina, USA) with a PE150 read length following the manufacturer's standard protocol. Low-quality reads, adapter-contaminated reads, and reads with >10 % unknown bases were filtered out to obtain clean reads. Clean reads were mapped to the mouse reference genome (GRCm39) via HISAT2 (v2.2.1), and gene expression levels were quantified as FPKM values. DEGs between groups were identified via DESeq2 (v1.34.0) with the following cutoffs: $\log_2(\text{fold change}) > 2$ and adjusted p value <0.05. The functional annotation of the DEGs was performed via Gene Ontology (GO) enrichment analysis and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis, with significance defined as an adjusted p value <0.05. The results were visualized via ggplot2 (v3.4.4) in R software (v4.2.3).

2.18. Immunoprecipitation coupled mass spectrometry (IP-MS) analysis

The IP protocol is described in section 2.10. The eluted proteins were reduced with 10 mM dithiothreitol, alkylated with 20 mM iodoacetamide, and digested with trypsin at 37 °C overnight. Tryptic peptides were analyzed via a Q Exactive HF-X mass spectrometer coupled with an Easy-nLC 1200 ultrahigh-performance liquid chromatography system. The MS1 scan range was 350–1800 m/z ; MS2 scans were performed with higher-energy collisional dissociation. Peptide and protein identification was performed via MaxQuant (v1.6) against the UniProt mouse protein database. The false discovery rate (FDR) for peptide and protein identification was set to <1 %. For protein quantification, the label-free quantification (LFQ) intensity was used, and interacting proteins of Mid1 were screened with the following cutoff: LFQ intensity ratio > 2 and p value <0.05.

2.19. GalNac-siMid1

GalNac-siMid1 was designed on the basis of Mid1 mRNA (GenBank: NM_001290504.1). Sense sequences: 5'-GUAACUCACCAUUCUUAU-3'; antisense sequences: 5'-AUAAGA UUGGUGAGAUUACUC-3'. siRNA strands were chemically synthesized by GenePharma (Shanghai, China) with 2'-O-methyl modification at the 3'-terminal nucleotides for nuclease resistance. Duplexes were purified by HPLC, and their integrity was verified via 15 % urea-PAGE. GalNac was conjugated to the 3'-end of the siRNA sense strand via a 6-carbon spacer phosphorothioate (PS) linker, enabling binding to the hepatocyte asialoglycoprotein receptor (ASGPR).

2.20. Statistical analysis

Statistical analyses were performed using two-tailed unpaired Student's t -tests or one-way ANOVA with Tukey's multiple comparisons test, based on the data distribution and experimental design. Specific statistical tests for each dataset are detailed in the corresponding figure legends. The experiments were repeated at least three times. Data processing was performed via GraphPad Prism 9.1 and SPSS 25.0 for analysis and visualization. A $*p < 0.05$ was considered statistically significant.

3. Results

3.1. E3 ubiquitin ligase Mid1 in the liver is elevated in both a mouse model of sepsis and infected patients

To investigate changes in ubiquitination in the livers of septic mice, we first assessed global ubiquitination levels. We found that livers from cecal ligation and puncture (CLP) mice exhibited significantly greater ubiquitination than did those from sham controls (Fig. 1A). The intravenous administration of the ubiquitination inhibitor TAK-243 decreased total ubiquitination by immunoblotting (Fig. 1B, S1A) and reduced the mRNA expression levels of the inflammatory factors TNF- α ,

IL-1 β and Tnfaip3 (TNF α -induced protein 3) according to RT-qPCR (Fig. S1B) in the livers of CLP mice but did not alleviate liver injury (Fig. S1C) or improve survival rates (Fig. 1C). To determine whether a specific ubiquitin ligase plays a dominant role, we performed transcriptomic analysis of CLP and sham mouse livers. A total of 393 DEGs were enriched in inflammatory response, immune regulation, and hepatocyte function pathways (Fig. 1D; Fig. S1D, E). Focusing on E3 ubiquitin ligases from the UBIBROWSER database, we identified three candidate E3 ubiquitin ligases (mid1, Pdzrn3, and Rnf180) (Fig. S1F). The heatmap shows DEGs of genes of partially interested in the livers of the Sham and CLP groups (Fig. 1E). A chord diagram showing the functional enrichment of these DEGs is shown in Fig. 1F. These results suggested that Mid1 presented the highest expression and pathway enrichment. Consistently, mid1 mRNA and protein (Fig. 1G-H) expression levels in the liver were elevated in both the LPS-induced sepsis model and the CLP-induced sepsis model. Moreover, liver samples from patients with acute cholangitis also presented increased MID1 mRNA (Fig. 1I) and protein expression (Fig. 1J-K, S1G). These findings suggest that mid1 may play a critical role in infection-related liver pathology.

3.2. Mid1 expression is predominantly elevated in hepatocytes from septic liver tissue

Although hepatocytes constitute the majority of liver cells, we sought to specifically determine whether Mid1 upregulation during sepsis occurs primarily in these cells. To investigate this, we isolated hepatocytes from CLP mice via gradient centrifugation, and separated them into hepatocyte and nonhepatocyte fractions (Fig. 2A). Successful separation was confirmed by analyzing hepatocyte-specific and inflammatory markers (Fig. 2B). The Mid1 mRNA and protein (Fig. 2B-C) levels were significantly greater in hepatocytes than in nonhepatocyte cells via quantitative analysis. We further validated these findings in different cell lines (not only primary hepatocytes) treated with LPS or infected with *E. coli* for 12 h, including hepatocytes (AML12), Kupffer cells (macrophages), liver sinusoidal endothelial cells (LSECs), and hepatic stellate cells (JS1), and the Mid1 protein (Fig. 2D) and mRNA (Fig. 2E) expression levels were predominantly elevated in hepatocytes. Primary hepatocytes were isolated from wild-type C57BL/6 J mice via in situ perfusion (Fig. S2A) and, after 24 h of culture, were treated with LPS or infected with *E. coli* for 12 h. Subsequent analyses revealed a differential response. While *E. coli* infection robustly increased both Mid1 mRNA (Fig. 2F) and protein levels (Fig. 2G-I); LPS stimulation resulted in only a marginal upregulation compared to the control. This disparity is likely attributable to the distinct signaling mechanisms engaged by purified LPS (primarily via TLR4) versus whole *E. coli*, which activates a broader repertoire of pattern recognition receptors. Collectively, these results demonstrate that Mid1 is upregulated in hepatocytes during bacterial infection, with a notably stronger induction by whole bacteria than by LPS alone.

3.3. Mid1 inhibition enhances hepatocyte antibacterial activity and reduces the degree of cellular damage induced by infection

To investigate the role of Mid1 in hepatocytes during infection, we generated murine Mid1-knockdown (shMid1) and Mid1-overexpressing (Lv-Mid1) AML12 hepatocyte cell lines, the efficiency of which was confirmed via RT-qPCR (Fig. 3A) and immunoblot analysis (Fig. 3B, C). Notably, under unstimulated conditions with low basal Mid1 expression, shRNA-mediated knockdown only slightly reduced Mid1 protein levels. In contrast, upon stimulation, the shRNA directly abrogated the robust, inducible synthesis of Mid1, resulting in a far more pronounced reduction in Mid1 abundance (Fig. 3B). Correspondingly, Lv-Mid1 cells exhibited further elevation of Mid1 protein levels following stimulation with LPS or *E. coli* (Fig. 3C). These observations led us to hypothesize that LPS or *E. coli* signaling either inhibits a pathway responsible for Mid1 degradation or promotes post-translational modifications that

enhance Mid1 stability, thereby driving its accumulation independent of its transcriptional regulation. These data suggest that Mid1 upregulation upon stimulation is achieved through a combination of transcriptional induction and potential protein stabilization. Supernatants from shMid1 cells collected 12 h post-infection (*E. coli* or MRSA) yielded significantly fewer bacterial colonies than shNC controls did in plating assays, whereas Lv-Mid1 supernatants produced more colonies (Fig. 3D, E). Consistent with these findings, live/dead bacterial staining revealed fewer live bacteria and a higher dead-to-live bacteria ratio in shMid1 supernatants but a lower ratio in the Lv-Mid1 groups (Fig. 3F, G). These results demonstrated that Mid1 suppression bolsters the antibacterial capacity of hepatocytes. Consistent with these findings, shMid1 AML12 hepatocytes presented fewer bacteria, but Mid1 overexpression (Lv-Mid1) AML12 hepatocytes presented more bacteria after incubation with *E. coli* for 24 h, according to microscopy images (Fig. 3H) and bacterial growth curve analysis (Fig. 3I). Trypan blue staining revealed significantly fewer dead cells in the shMid1 groups than in the control or Lv-mid1 groups post-infection (Fig. 3H, J). Flow cytometry (annexin V-PE/7-AAD) confirmed these results (Fig. 3K). These results indicate that Mid1 inhibition decreases the number of bacterial colonies and mitigates infection-induced hepatocyte damage in cells.

3.4. Hepatocyte-specific Mid1 knockdown improves organ status and survival in septic mice

To investigate the role of Mid1 in sepsis, we performed in vivo experiments using a liver-targeted adeno-associated virus (AAV) to construct a hepatocyte-specific Mid1 knockdown mouse model (Mid1^{hep-KD}) (Fig. 4A). RT-qPCR confirmed the knockdown efficiency (Fig. 4B), and fluorescence imaging of multiple organs demonstrated AAV-specific liver tropism (Fig. 4C). Compared with the AAV-NC group, the Mid1^{hep-KD} group presented a significantly greater survival rate after cecal ligation and puncture (CLP) or intraperitoneal *E. coli* infection (Fig. 4D). Additionally, the AST (aspartate aminotransferase), Cr (creatinine) and LDH (lactate dehydrogenase) activities were significantly lower in the Mid1^{hep-KD} group than in the control group, but the number of PLTs (platelets) was significantly greater (Fig. 4E, Fig. S2B). Blood PLTs are key mediators of inflammation and infection [24]. These results suggested that Mid1 blockade alleviated liver and kidney injury induced by CLP. Notably, bacterial loads in major organs were markedly reduced in Mid1^{hep-KD} mice (Fig. 4F, G).

Histopathological analysis of liver tissue revealed extensive immune cell infiltration in both the Mid1^{hep-KD} and AAV-NC groups after CLP or *E. coli* challenge. However, compared with control mice, Mid1^{hep-KD} group mice presented significantly smaller areas of hepatocyte degeneration and necrosis (orange arrows) and less immune cell infiltrates (green arrows) in H&E-stained liver sections (Fig. 4H) and a lower histological activity index (Fig. 4I). We further assessed inflammatory responses in peritoneal macrophages. The mRNA expression levels of proinflammatory factors (TNF α , IL-1 β and TGF β) and the key transcription factor STAT1 were significantly decreased after CLP. However, the mRNA expression level of CD206, a marker of anti-inflammatory M2 macrophages, was lower in the Mid1^{hep-KD} group than in the AAV-NC group, suggesting a greater risk of cytokine storm in the AAV-NC group (Fig. S2C). In summary, hepatocyte-specific Mid1 knockdown improves organ function, reduces bacterial dissemination, and enhances survival in septic mice, potentially by mitigating excessive inflammation.

3.5. HRG is a substrate of Mid1-mediated ubiquitination

To further explore the mechanism of Mid1 function, we performed immunoprecipitation (IP) using an anti-Mid1 antibody in liver lysates from CLP and sham mice, followed by mass spectrometry analysis (IP-MS) to identify Mid1-binding proteins, particularly those that were differentially expressed between the two groups. The results revealed

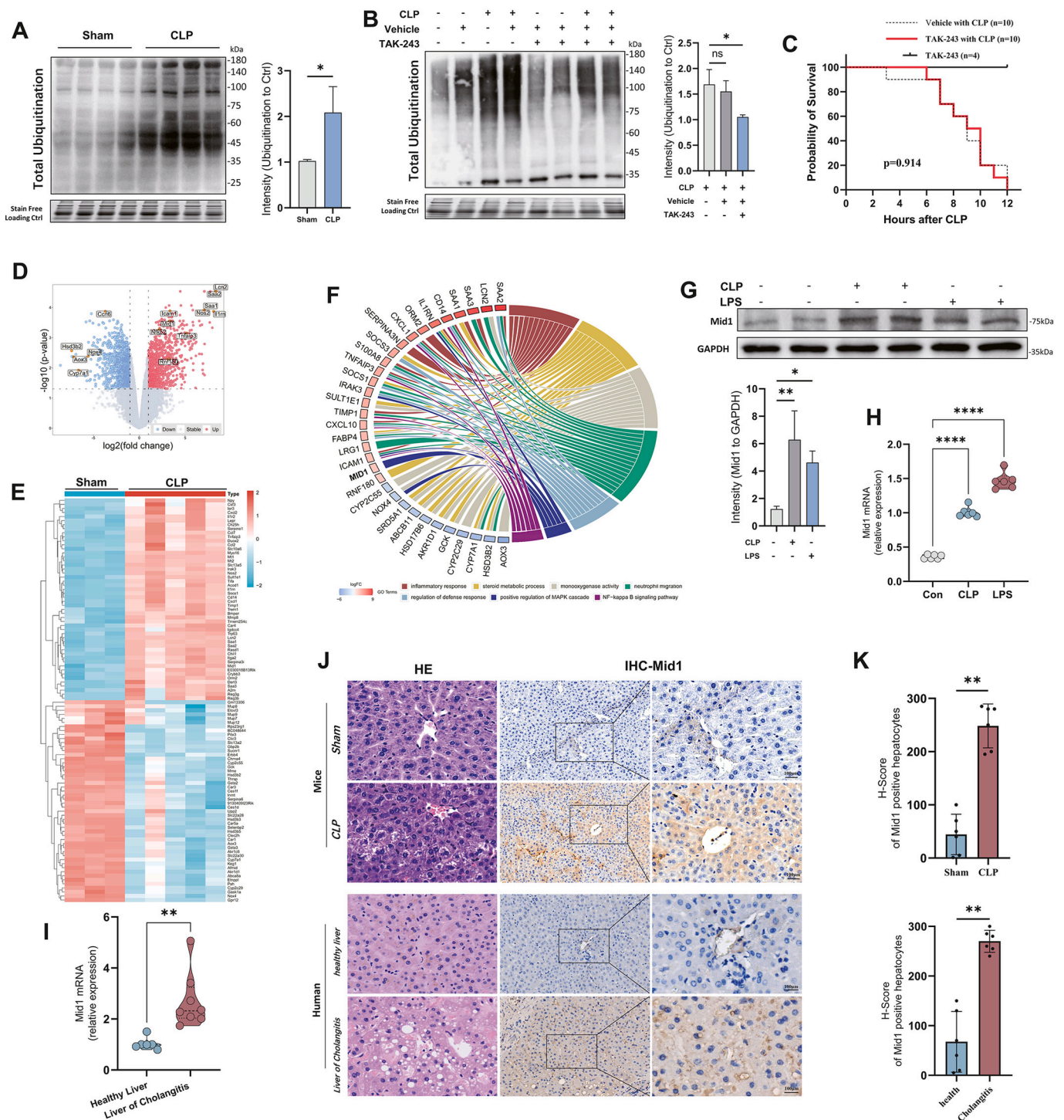


Fig. 1. The level of Mid1 in the liver, which is associated with inflammation, was significantly elevated. (A) Immunoblot analysis of overall ubiquitination levels in the livers of mice 12 h after CLP; Sham surgery was used as a control. The results of the immunoblot analysis of 3 biological replicates are shown alongside (mean with SEM, * $p < 0.05$, paired Student's t -test). (B) Immunoblot analysis of the overall ubiquitination level in livers from mice after 20 mg/kg TAK-243 or vehicle treatment (t.v.i.) 24 h before CLP. The results of the immunoblot analysis of 3 biological replicates are shown alongside (mean with SEM, * $p < 0.05$, one-way ANOVA). (C) Survival of CLP mice after Tak-243 treatment (t.v.i.) for 24 h. Survival data are shown as Kaplan–Meier curves. (D) Volcano map of differentially expressed genes in the livers of CLP and control mice. Some of the top 50 genes (including Mid1) are shown. (E) Heatmap of differential gene expression in livers from CLP and control mice. (F) Chord diagram showing the functional enrichment of the 393 DEGs. The intensity of the red and blue squares represents the magnitude of the difference between them. (G) Immunoblot analysis of Mid1 in the liver 12 h after intraperitoneal injection of LPS or CLP. Quantification of the results from 3 biological replicates is shown (mean with SEM, * $p < 0.05$, ** $p < 0.01$; one-way ANOVA). (H) RT-qPCR of Mid1 in the liver 12 h after the intraperitoneal injection of LPS or CLP (**** $p < 0.0001$, one-way ANOVA). (I) RT-qPCR analysis of Mid1 mRNA levels in healthy and cholangitis liver samples from patients (** $p < 0.01$, unpaired Student's t -test). (J) Hematoxylin-eosin staining (left) and immunohistochemistry of Mid1 (right) in livers from mice and humans. (K) H-scores of Mid1-positive hepatocytes in mice and humans from (J) (mean with SD, ** $p < 0.01$, Mann-Whitney test).

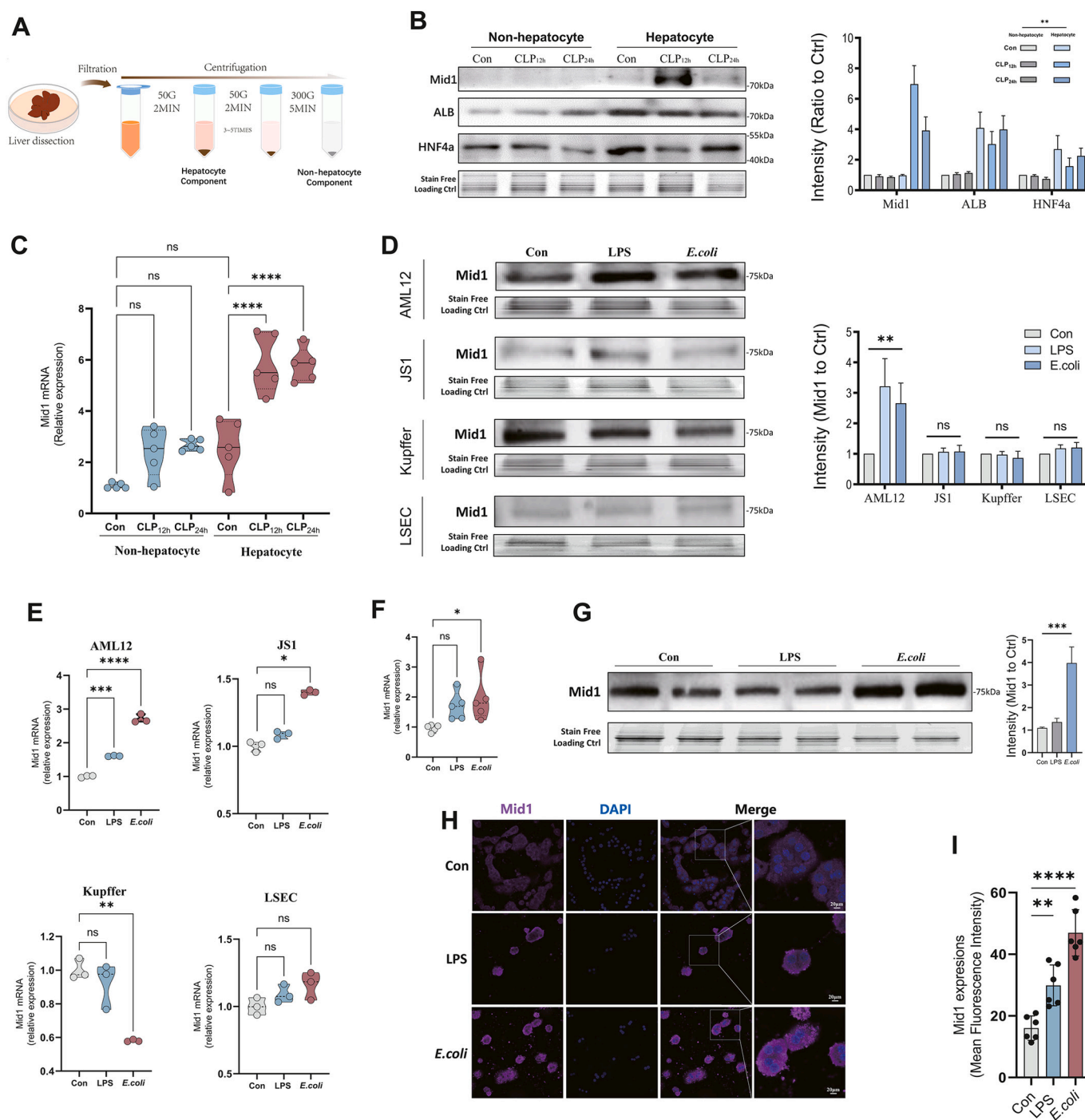


Fig. 2. Mid1 upregulation in the septic liver occurs predominantly in hepatocytes. (A) Experimental workflow for hepatocyte/nonhepatocyte separation via gradient centrifugation. (B) Immunoblot analysis of the protein levels of ALB, HNF4 α , and Mid1 in the hepatocyte and nonhepatocyte fractions at 12 h and 24 h post-CLP. The results of the immunoblot analysis of 3 biological replicates are shown alongside (mean with SEM, $^{**}p < 0.01$, paired Student's *t*-test). (C) RT-qPCR analysis of Mid1 mRNA levels in hepatocyte and nonhepatocyte fractions at 12 h and 24 h post-CLP. (D) Immunoblot analysis of Mid1 protein expression in different liver cell lines (AML12 hepatocytes, JS1 stellate cells, Kupffer macrophages, and LSECs) following LPS treatment or *E. coli* infection. The results of the immunoblot analysis of 3 biological replicates are shown alongside (mean with SEM, $^{*}p < 0.01$, one-way ANOVA). (E) RT-qPCR analysis of Mid1 mRNA expression in liver cell lines after LPS treatment or *E. coli* infection for 12 h ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$; one-way ANOVA). (F) Mid1 mRNA expression in primary mouse hepatocytes after LPS treatment or *E. coli* infection for 12 h, as determined by RT-qPCR ($^{**}p < 0.01$, $^{****}p < 0.0001$, one-way ANOVA). (G) Immunoblot analysis of Mid1 protein levels in primary mouse hepatocytes after LPS treatment or *E. coli* infection for 12 h. Quantification of 3 biological replicates is shown (mean with SD, $^{***}p < 0.001$, one-way ANOVA). (H) Mid1 expression in primary mouse hepatocytes after LPS treatment or *E. coli* infection for 12 h, as determined by immunofluorescence. (I) Statistical analysis of the intensity of Mid1 staining in (H) (means with SDs, $^{*}p < 0.05$; one-way ANOVA).

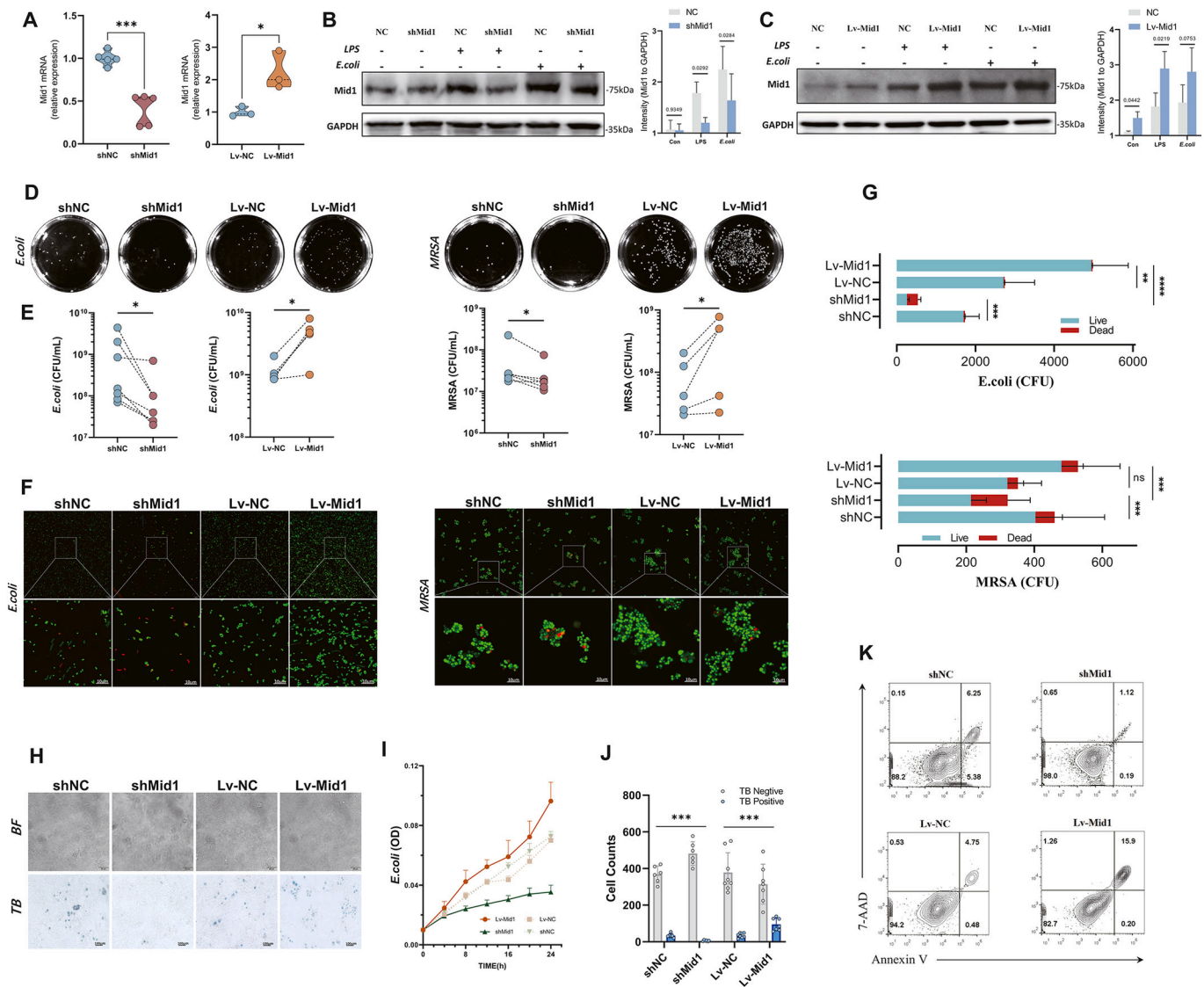


Fig. 3. Mid1 regulates antibacterial capacity and resistance to infection in hepatocytes. (A) RT-qPCR analysis of relative Mid1 mRNA levels in shMid1 and Lv-Mid1 cell lines compared with their respective controls (**p* < 0.05, ****p* < 0.001, unpaired Student's *t*-test). (B–C) Immunoblot analysis of Mid1 protein expression in shMid1 and Lv-Mid1 cell lines following LPS treatment or *E. coli* infection for 12 h. Quantification of 3 biological replicates is shown (means with SDs, multiple paired *t*-tests). (D) Representative images of bacterial colonies in cell supernatants collected 12 h post-infection (*E. coli* or MRSA). (E) Quantification of colony-forming units (CFUs) from (D). The connected dots indicate paired experimental replicates (**p* < 0.05, paired Student's *t*-test). (F) Live/dead bacterial staining (confocal microscopy) of supernatants 12 h post-infection. Green: viable bacteria (SYTO9); red: dead bacteria (propidium iodide). (G) Quantification of bacterial viability from (F) (*n* = 3 fields per group; ***p* < 0.01, ****p* < 0.001, chi-square test). (H) Phase-contrast microscopy and trypan blue staining of AML12 hepatocytes 24 h post-*E. coli* infection. (I) Bacterial growth kinetics in cell lines with differential Mid1 expression. (J) Trypan blue-positive cell counts from H (*n* = 6 fields per group; ****p* < 0.001, chi-square test). (K) Apoptosis/necrosis analysis by flow cytometry (annexin V-PE vs. 7-AAD). Quadrant II: dead cells.

that the capping protein Capza1, antimicrobial protein HRG, and other molecules were potential Mid1 interacting proteins (Table 1). Transcriptome differential analysis of livers from Mid1^{hep-KD} and NC mice 12 h after CLP was performed. According to the GO/KEGG pathway enrichment results, the response to lipopolysaccharide, response to chemokines, positive regulation of cytokine production and heparin binding were identified (Fig. 5A). The molecular function enrichment of the HRG PPI network associated with STING was also enriched in heparin binding, signaling receptor binding and lipoprotein lipase activator activity (Fig. 5B). The protein-protein interaction (PPI) network of HRGs, the biological process (BP) and cellular component (CC) enrichment of the HRG network were shown in Fig. S3a, 3b, and 3c, respectively. The heparin-binding and response to lipopolysaccharide pathways, both closely associated with HRG function, were significantly enriched, suggesting a potential link between Mid1 and HRG. To

determine whether Mid1 interacts with HRG, native PAGE, immunoblotting and coimmunoprecipitation analyses were applied. Native PAGE permits high-resolution separation of multiprotein complexes under native conditions to illustrate protein-protein interaction [25]. Mid1 (Fig. 5C left) and HRG (Fig. 5C middle) were visualized almost exclusively from multiprotein complexes of the same molecular weight (>310 kDa) by immunoblotting. Blue native PAGE revealed the multiprotein complexes of whole cell lysates (Fig. 5C right). While these results indicated the presence of protein complexes containing various proteins, likely including Mid1 and HRG, they did not prove a direct interaction between them. To investigate this, we performed coimmunoprecipitation assays, which confirmed that Mid1 interacts with HRG (Fig. 5D). Furthermore, immunofluorescence analysis demonstrated that Mid1 and HRG colocalized in the liver (Fig. 5E, S4a, b), and this colocalization exhibited a negative correlation under CLP-

induced sepsis-like conditions (Fig. 5F). In addition, hepatocytes exhibited increased ubiquitination levels and elevated Mid1 expression in CLP model mice; whereas HRG levels decreased in both hepatocytes and serum (Fig. 5G). To determine whether Mid1 regulates HRG abundance, we conducted *cellulo* experiments and showed that Mid1 inhibition in hepatocytes significantly increased the levels of intracellular HRG and its secretion into culture supernatants after infection (Fig. 5H). Similarly, *in vivo* experiments revealed higher serum and liver HRG levels in Mid1^{hep-KD} mice than in NC mice postsepsis induction (Fig. 5I). To further investigate the underlying mechanism, we used the proteasome inhibitor MG132 to evaluate whether Mid1 mediates HRG ubiquitination. Compared with untreated control hepatocytes, AML12 shMid1 hepatocytes stimulated with the proteasome inhibitor MG132 presented increased intracellular HRG levels and ubiquitin-bound protein content according to immunoblotting (Fig. 5J left). Ubiquitin immunoprecipitation assays further demonstrated that HRG-ubiquitin binding was markedly reduced upon Mid1 knockdown (Fig. 5J right), confirming that the E3 ubiquitin ligase Mid1 mediates HRG ubiquitination. The quantification revealed a significant difference (Fig. 5K). These results suggest that HRG undergoes ubiquitin-dependent degradation following sepsis. Moreover, treatment with the protein synthesis inhibitor cycloheximide (CHX) revealed that Mid1 overexpression (Lv-Mid1) markedly reduced HRG protein levels in AML12 cells, whereas Mid1 knockdown (shMid1) had no notable effect (Fig. 5L). Critically, the degradation of HRG in Mid1-overexpressing cells treated with CHX was effectively blocked by the proteasome inhibitor MG132 (Fig. 5L, M, S4c), demonstrating that Mid1 promotes HRG degradation via the ubiquitin-proteasome pathway. In summary, these findings demonstrate that HRG serves as a substrate for Mid1-mediated ubiquitination in hepatocytes.

3.6. HRG antagonization attenuates the antibacterial phenotype of shMid1 cells

As shown in Fig. S2C, inflammatory factor expression levels in peritoneal macrophages from Mid1^{hep-KD} mice were altered. To determine whether Mid1 knockdown modulates immune cell activity via HRG, we performed immunofluorescence colocalization of the hepatic HRG with T cells (CD3⁺), neutrophils (CD15⁺), and macrophages (CD68⁺) in Mid1^{hep-KD} mice after CLP. The results revealed significant interactions between HRG-CD68⁺ macrophages, which were further enhanced in Mid1^{hep-KD} mice. Additionally, CD15⁺ neutrophil infiltration in the liver was markedly lower in Mid1^{hep-KD} mice than in control mice (Fig. 6A, B). Mid1 has multiple substrates, including the well-characterized protein PP2Ac, which was also upregulated in hepatocytes from CLP mice (Fig. 6C) and *E. coli*-infected shMid1 cells (Fig. 6D). To determine the role of HRG in Mid1-mediated antimicrobial effects, we treated *E. coli*-infected shMid1 cells with 400 µg/ml heparin, a competitive HRG antagonist. Immunoblot analysis revealed that heparin reduced HRG binding to bacteria, resulting in more HRG remaining in the supernatant (Fig. 6E). Bacterial plating assays confirmed that heparin treatment significantly suppressed the increased antibacterial activity of shMid1 cells (Fig. 6F, G, S5A) and increased cell death (Fig. 6H). However, CCK-8 cytotoxicity assays (Fig. S5B) and bacterial PI staining (Fig. S5C) also demonstrated that this heparin concentration did not affect cell or bacterial viability. In contrast, clinical heparin doses (1–10 µg/ml) failed to antagonize HRG or reverse the phenotype (Fig. S5D), underscoring the specificity of the role of HRG. These findings suggest that hepatocytes regulate antibacterial activity through the Mid1-HRG axis.

3.7. Hepatocyte-specific Mid1 siRNA and DPP4 inhibitors improve the systemic status and survival of CLP mice

Using the GalNAc-siRNA conjugate delivery platform, we developed a hepatocyte-targeted GalNAc-siMid1 and administered it intravenously

24 h prior to CLP induction (Fig. 7A). Fluorescence imaging of organs 12 h post-CLP confirmed the liver-specific knockdown efficiency of GalNAc-siMid1 (Fig. 7B), which was further validated by RT-qPCR (Fig. 7C). Compared with CLP control mice, GalNAc-siMid1 mice presented significantly lower bacterial loads in multiple organs (Fig. 7D, E) and improved serological marker and HRG concentrations (Fig. 7F, S6A). Immunoblot analysis revealed elevated levels of hepatocyte-derived HRGs in both the serum and liver tissue of the GalNAc-siMid1 group (Fig. 7G). Given recent evidence that dipeptidyl peptidase-4 (DPP4) knockout suppresses Mid1 expression [26], we evaluated the therapeutic potential of the oral DPP4 inhibitor linagliptin. In CLP mice, linagliptin administration modestly reduced hepatic Mid1 mRNA levels compared with saline treatment (Fig. 7H, I). It did not affect glycemic fluctuations, as measured by continuous glucose monitoring (Fig. 7J), but increased overall survival (Fig. 7K). These findings demonstrate that GalNAc-siMid1 may serve as promising therapeutic agents for sepsis by targeting hepatic Mid1.

3.8. Discussion

Previous studies have revealed that hepatocytes, the major parenchymal cells in the liver, maintain the homeostasis of inflammation [27] and host–pathogen interactions, such as those involving HIV [28], HBV [29], and HCV [30]. However, there is limited evidence suggesting that hepatocytes can directly combat bacterial infections. This study aimed to investigate the role and underlying mechanism of the hepatocyte ubiquitin E3 ligase *Mid1* in regulating antibacterial defense during sepsis.

Hepatocytes produce various immune-modulating factors, including C-reactive protein (CRP), serum amyloid A (SAA), LPS-binding protein (LBP), CD14, MD-2, fibrinogen [31], hepatocyte growth factor (HGF) [32], and lipocalin-2 (LCN2) [33], to activate the host response to control bacterial infections. Additionally, primary mouse hepatocytes enhance the type I interferon response, whereas HepG2 cells induce a type III interferon response to defend against *Listeria monocytogenes* infection. Bacterial infections also alter the expression of genes associated with cholesterol metabolism in HepG2 and Huh7 cells [34]. Studies in mice have shown that the MyD88 pathway in hepatocytes infected with lethal *Ehrlichia* triggers CASP11 activation, gasdermin D cleavage, IL-6 production, and inhibition of bacterial proliferation [35]. In addition to their immune functions, hepatocytes contribute to metabolic adaptation during infections by enhancing gluconeogenesis and fatty acid β-oxidation, processes closely linked to sepsis prognosis [36,37]. Hepatocytes also engage in cross-talk with other immune cells during bacterial infections. For example, in turtles infected with *Aeromonas hydrophila*, hepatic tissues presented increased numbers of endothelial and Kupffer cells but reduced numbers of T cells, NK cells, dendritic cells, and erythroid cells [38]. Despite these findings, research on hepatocyte-mediated infection responses remains limited compared with the extensive studies on the anti-inflammatory and immunomodulatory mechanisms of immune cells [39]. These results suggest that the potential roles of hepatocytes in bacterial infection, underscoring the need for further therapeutic strategies for septic liver injury.

Our transcriptomic analysis revealed that *mid1* is upregulated in the liver during sepsis. Initially, identified as a microtubule-associated protein, *mid1* was linked to Opitz syndrome through genetic mutations [40]. Subsequent studies have demonstrated its involvement in immune regulation, including T-cell migration in atherosclerosis [26] and effector T-cell trafficking in experimental autoimmune encephalomyelitis (EAE) [17]. Some studies have defined Mid1 as a transcription factor [41]. MID1 binds to *ATXN2*, *ATXN3*, and *ATXN7* mRNAs to promote protein synthesis in a CAG repeat-containing mRNA-dependent manner [42]. Our unpublished results suggest that HRG transcription in hepatocytes from CLP mice is suppressed [9]. However, the role of Mid1 in liver-related diseases and infections remains unclear and warrants further investigation. Only a few studies have identified Mid1 as a

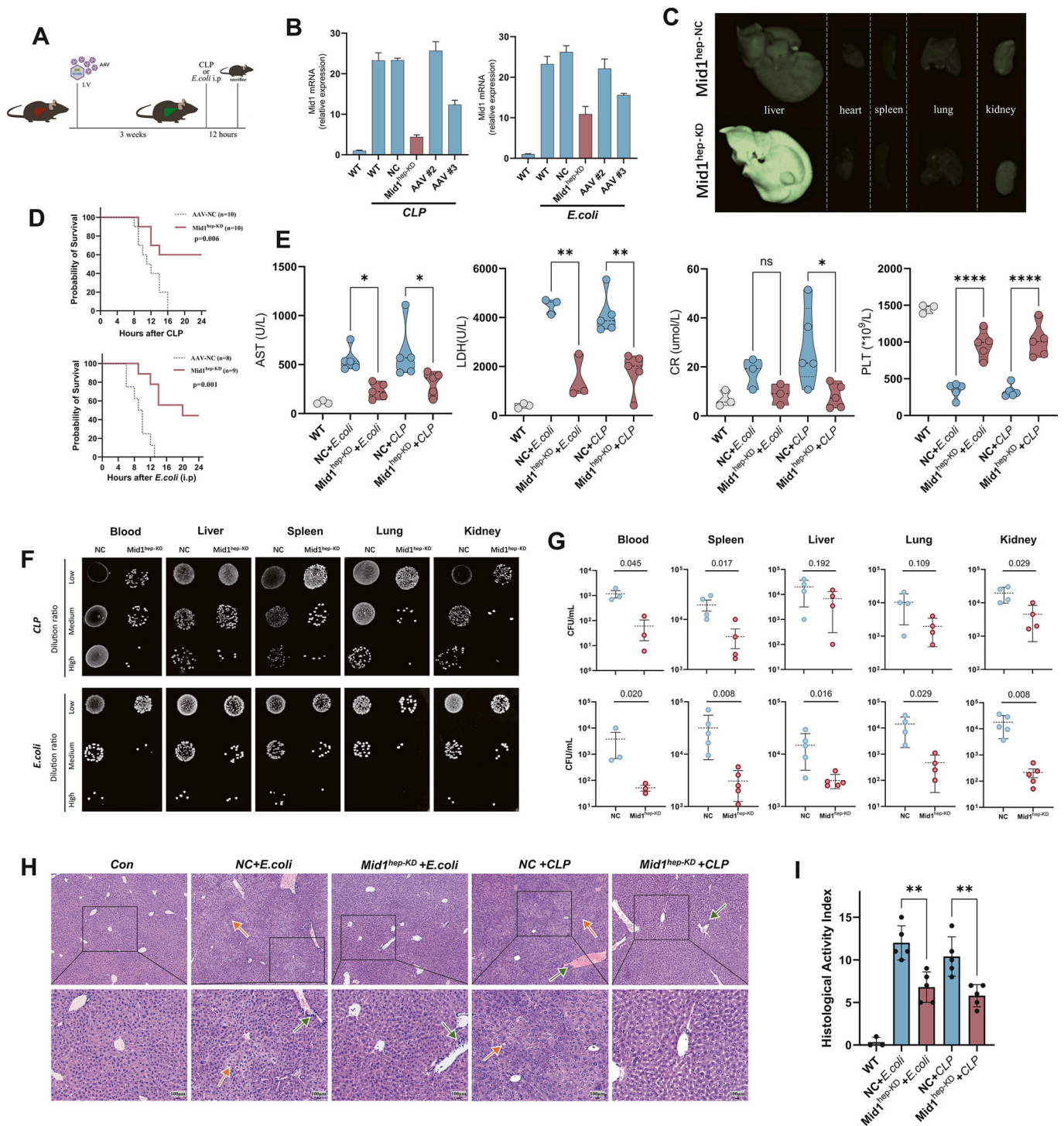


Fig. 4. Hepatocyte-specific *Mid1* knockdown enhances sepsis outcomes. (A) Schematic of the experimental design for generating *Mid1^{hep-KD}* mice. (B) Relative *Mid1* knockdown efficiency in livers from *Mid1^{hep-KD}* mice 12 h post intraperitoneal injection of *E. coli* or CLP. (C) Fluorescence imaging (EGFP) of organs from *Mid1^{hep-KD}* and negative control (NC) mice. (D) Survival analysis of *Mid1^{hep-KD}* and NC mice following *E. coli* challenge or CLP. (E) Serum biochemical marker levels in *Mid1^{hep-KD}* and NC mice at 12 h post-infection or CLP. ($p < 0.05$, $**p < 0.01$, $***p < 0.0001$; unpaired Student's *t*-test or Mann-Whitney *U* test). (F) Bacterial colony growth from organ lysates. Whole organs were homogenized, and lysates were plated at high, medium, and low dilutions. (G) Quantification of bacterial colony-forming units (CFUs). Each dot represents one mouse. (p -values were determined by unpaired Student's *t*-test or Mann-Whitney *U* test). (H) H&E-stained liver sections from infected *Mid1^{hep-KD}* and NC mice. Degenerated/necrotic hepatocytes (orange arrows) and immune cell infiltrates (green arrows) are indicated. (I) Histological activity indices (Knodell scores) from (H) (means with SDs, $**p < 0.01$; one-way ANOVA). NC, negative control.

significant factor in the pathogenesis of liver disease. *Mid1* is upregulated in Kupffer cells during hepatic ischemia-reperfusion injury (IRI). Inhibition of *Mid1* attenuates liver damage by reducing neutrophil infiltration, intrahepatic inflammation, and hepatocyte apoptosis [43].

Additionally, *mid1* modulates endothelial cell ICAM-1 expression and neutrophil adhesion in the lungs of septic mice [15].

Mid1 regulates the innate immune response by ubiquitinating key signaling molecules, including IRF3 [18], IFNAR2 [44], and f NSP9

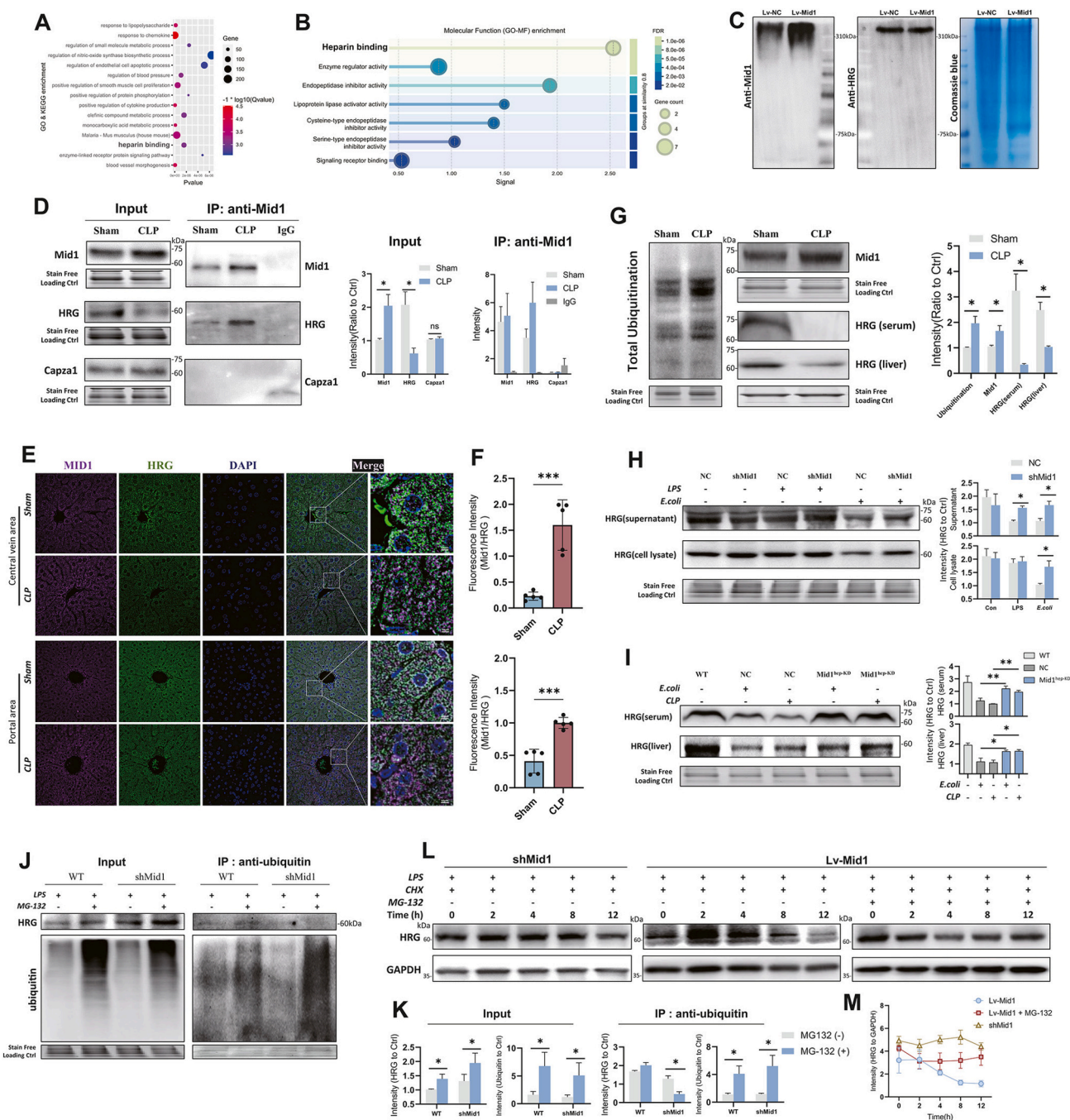


Fig. 5. Mid1 mediates the ubiquitination-dependent degradation of HRG in hepatocytes. (A) GO/KEGG pathway enrichment analysis of differentially expressed genes in Mid1^{hep-KD} versus NC mice 12 h after CLP. (B) Molecular function enrichment of the HRG PPI network by STING (<https://cn.string-db.org>). (C) Native gel immunoblot analysis of Mid1 and HRG levels in Lv-Mid1 (mid1 overexpression) and Lv-NC hepatocytes after *E. coli* infection. Protein complexes containing Mid1 or HRGs of the same molecular size. (D) Left: Immunoblot analysis of Mid1 (positive control), HRG, and Capza1 (negative control) protein levels in liver tissues from CLP and sham-operated mice. Right: Coimmunoprecipitation (Co-IP) assay using an anti-Mid1 antibody or an isotype IgG control. The results of the immunoblot analysis of 3 biological replicates are shown alongside (mean with SEM, $p < 0.05$, multiple paired *t*-tests). (E) Laser confocal microscopy images of mouse liver sections stained with anti-Mid1 and anti-HRG antibodies. (F) Statistics of the relative fluorescence intensity of Mid1/HRG from (E) (means with SDs, $***p < 0.001$, unpaired *t*-test). (G) Immunoblot analysis of total ubiquitinated proteins, Mid1, and HRG in liver tissues, along with serum HRG levels, in CLP- and sham-operated mice. The results of the immunoblot analysis of 3 biological replicates are shown alongside (means with SDs, $p < 0.05$, multiple paired *t*-tests). (H) Immunoblot analysis of HRG levels in supernatants and cell lysates from shMid1 (mid1 knockdown) and NC cell lines 12 h after LPS treatment or *E. coli* infection. The results of the immunoblot analysis of 3 biological replicates are shown alongside (mean with SD, $p < 0.05$, Multiple paired *t*-tests). (I) Immunoblot analysis of liver and serum HRG levels in Mid1^{hep-KD} and NC mice 12 h after intraperitoneal *E. coli* injection or CLP. The results of the immunoblot analysis of 3 biological replicates are shown alongside (means with SDs, $p < 0.05$, $**p < 0.01$; one-way ANOVA). (J) Left: HRG and ubiquitin-binding protein levels in shMid1 and AML12 control cell lysates after MG132 treatment for 12 h (10 μ M). Right: Co-IP assay using an anti-ubiquitin antibody. (K) Quantification of (J) 3 biological replicates (means with SDs, $p < 0.05$, $**p < 0.01$; one-way ANOVA). (L) Left: HRG levels in shMid1 cell lysates at 0, 2, 4, 8, and 12 h after CHX (30 μ M) treatment. Right: HRG levels in Lv-Mid1 cell lysates after CHX (30 μ M) treatment or CHX + MG132 (10 μ M) cotreatment. (M) Quantification of (L) from 3 biological replicates (means with SDs). NC, negative control.

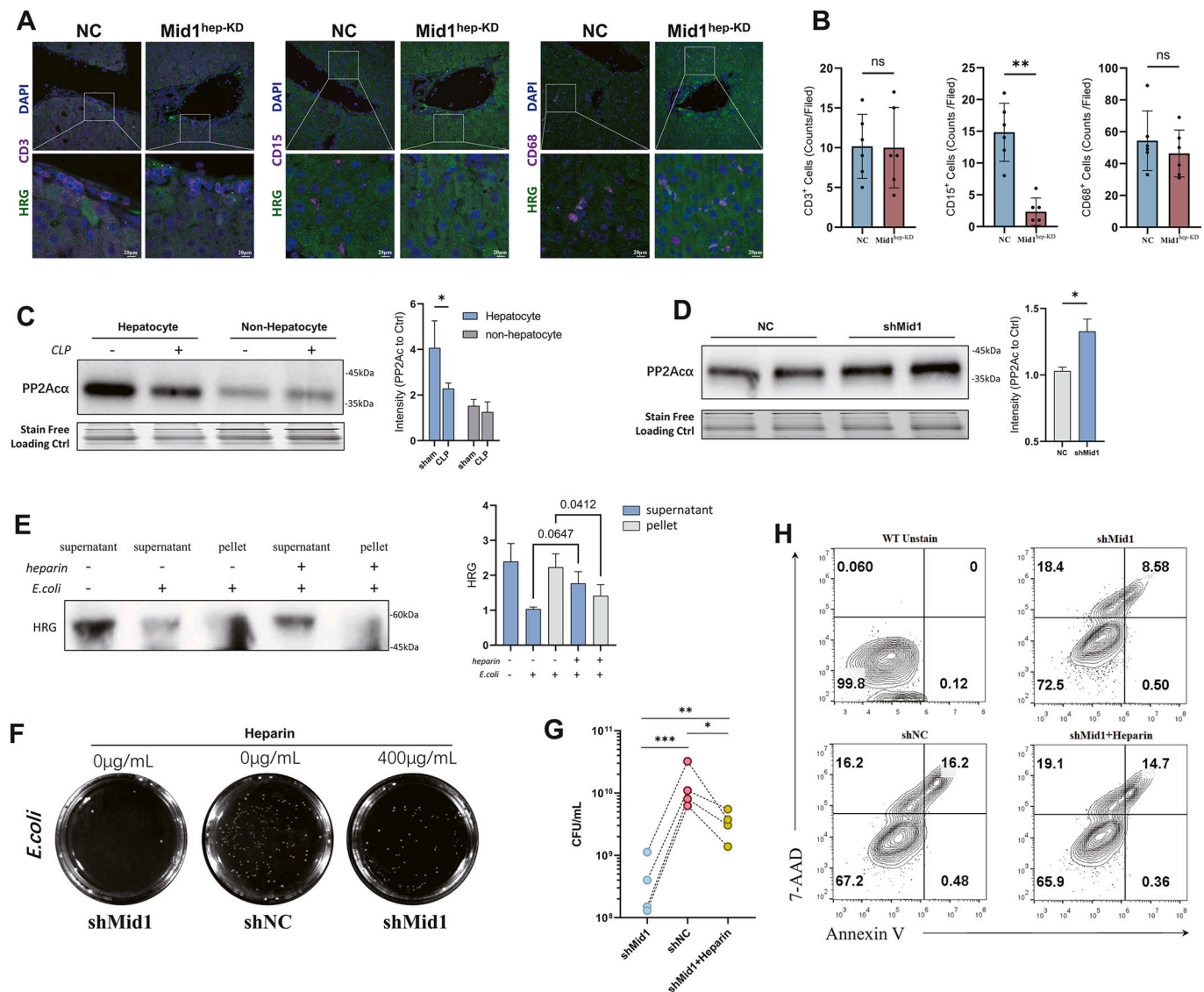
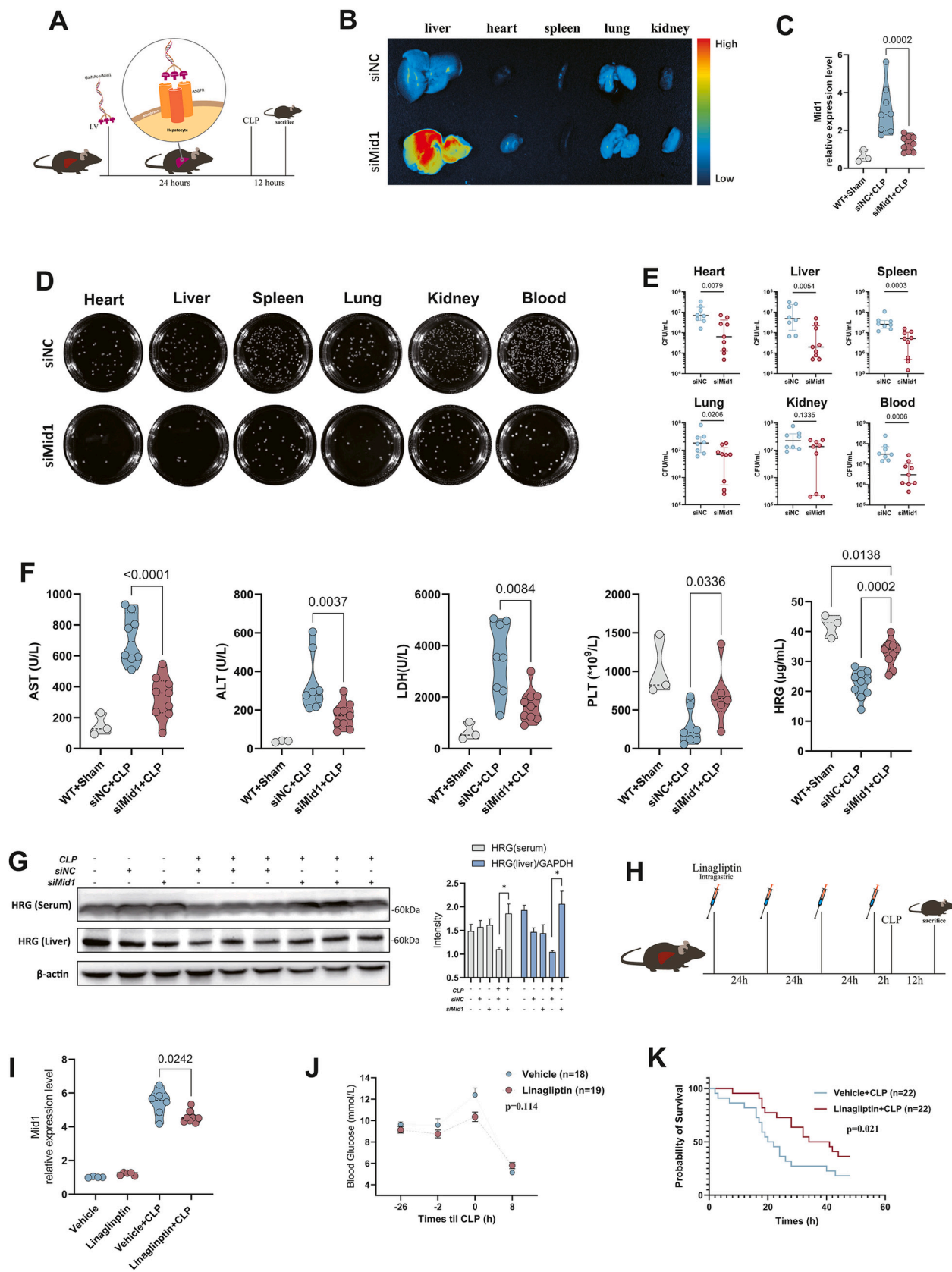


Fig. 6. Antagonizing HRG attenuated the antibacterial effect of shMid1. (A) Immunofluorescence imaging of immune cells in the livers of Mid1^{hep-KD} and NC mice 12 h after CLP. (Left) The infiltration of CD3⁺ T cells, was not significantly different between the groups. (Middle) Increased infiltration of CD15⁺ cells (primarily neutrophils) in NC mice. (Right) Comparison of CD68⁺ macrophage infiltration between the groups. Notably, HRGs exhibit significant cross-talk with macrophages. (B) Positive cell count from (A) (means with SDs, $^{**}p < 0.01$, Mann-Whitney test). (C) PP2Ac protein levels in hepatocytes and nonhepatocytes treated with or without CLP. The quantification of 3 biological replicates is shown alongside (means with SDs, $^{*}p < 0.05$, unpaired *t*-tests). (D) PP2Ac protein levels in shMid1 and NC cells after *E. coli* infection for 12 h. The quantification of 3 biological replicates is shown alongside (mean with SEM, $^{*}p < 0.05$, unpaired *t*-tests). (E) Effects of heparin treatment (400 μ g/ml, 30 min) on HRG binding to bacteria. The cell culture supernatant from *E. coli*-infected shMid1 cells was centrifuged and separated into supernatant and pellet fractions. Immunoblot analysis revealed that heparin reduced HRG binding to bacteria, resulting in more HRG remaining in the supernatant. The quantification of 3 biological replicates is shown alongside (means with SDs, one-way ANOVA). (F) Bacterial plating photos of cell supernatants 12 h after *E. coli* infection. (G) Quantification of colony counts from (F). The connected dots represent independent experiments. Significance was determined by paired Student's *t*-tests ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$). (H) Flow cytometry analysis (annexin V vs. 7-AAD) of infected cell lines. Quadrant II indicates the proportion of dead cells. NC, negative control.

[45], as well as by modulating PP2A through yeast two-hybrid analyses [13] during inflammation and infection. In addition, it has been shown to catalyze different ubiquitin chains, including K48- and K63-linked chains [46]. MARCH7 induces the degradation of target proteins through K48-linked ubiquitination. Conversely, the NLRP3 inflammasome is activated through K63-linked ubiquitination [47]. Several studies have demonstrated that the ubiquitin system plays a role in sepsis pathogenesis. CD36 in hepatocytes promotes the UBQLN1-dependent degradation of autophagic SNARE proteins, thereby exacerbating septic liver injury [48]. TRIM47 acts as a novel endothelial activation factor that exacerbates LPS-induced acute lung injury in mice through K63-linked ubiquitination of TRAF2 [49]. MARCH3 suppresses

the IL-3-triggered inflammatory response by promoting K48-linked polyubiquitination and subsequent degradation of IL-3R α [50]. MARCH proteins also regulate immune system [51]. However, our data revealed the specific role and underlying ubiquitin mechanisms of Mid1 in the progression of septic liver injury.

To further explore the mechanisms by which Mid1 regulates septic liver injury, we identified HRG as a novel ubiquitin-degrading substrate of Mid1 via immunoprecipitation and mass spectrometry analysis. Co-immunoprecipitation and proteomic analysis were also used to identify DPP4 as a candidate Mid1 substrate in MH7A, a human synovial cell line [52]. PP2Ac, recognized as a ubiquitination substrate of Mid1, was initially identified through yeast two-hybrid analyses [40]. A yeast



(caption on next page)

Fig. 7. GalNAc-siMid1 and linagliptin target Mid1 for sepsis therapy. (A) Schematic of GalNAc-siMid1 treatment (4 µg/g). (B) Fluorescence imaging (CY3) of organs from siMid1-treated and NC mice. (C) RT-qPCR analysis of Mid1 mRNA levels in livers from siMid1 and NC mice 12 h after CLP (one-way ANOVA). (D) Bacterial colony growth from plated whole-organ lysates from siMid1 and NC mice 12 h after CLP. (E) Quantification of colony-forming units (CFUs). Each dot represents one mouse (Mann-Whitney test). (F) Serological biomarkers in siMid1 and NC mice 12 h post-CLP. *P*-values were calculated via unpaired Student's *t*-test or the Mann-Whitney test (one-way ANOVA). (G) Immunoblot analysis of HRG levels in serum and liver tissue from siMid1 and NC mice after CLP for 12 h. The quantification of 3 biological replicates is shown alongside (mean with SEM, **p* < 0.05, Multiple paired *t*-tests). (H) Schematic of the linagliptin treatment regimen (80 µg/kg, once daily). (I) RT-qPCR analysis of Mid1 in livers from linagliptin-treated and NC mice from (H). Each dot represents one mouse (unpaired *t*-tests). (J) Continuous glucose monitoring. The data points represent the mean blood glucose levels. No significant difference was observed (*P* = 0.114, repeated-measures ANOVA). (K) Kaplan-Meier survival analysis of linagliptin-treated and NC mice after CLP. Linagliptin significantly improved survival (*P* = 0.021, Gehan-Breslow-Wilcoxon test). NC, negative control.

Table 1
Differential Mid1-binding proteins between sham and CLP mice liver by immunoprecipitation with mass spectrum (IP-MS).

Gene names	MW [kDa]	Protein score	Sequence coverage (%)	Unique Peptides	Peptides	PSMs	Abundances
Capza1	32.9	107	10	1	2	2	9.75e+05
Hrg	59.1	44	3	1	1	1	1.72e+06
Usp14	56	32	1	1	1	1	2.68e+06
Ptpmt1	21.9	31	7	1	1	1	1.72e+05
Irgm1	46.5	27	3	1	1	1	7.81e+05
Ndufa13	16.8	27	8	1	1	1	3.51e+05
Lsm4	15.1	26	5	1	1	1	7.45e+05
Usp24	293.8	25	0	1	1	1	1.20e+06
Ccdc82	61	24	1	1	1	1	9.13e+05
Dnajc11	63.2	22	1	1	1	1	1.16e+05

two-hybrid screening approach was employed to identify the MIG12 protein, which interacts with Mid1 to stabilize microtubules. This interaction was subsequently confirmed via immunoprecipitation techniques [53]. HRG, a hepatocyte-derived serum protein, has multiple functions, including immune modulation, anti-inflammatory effects, and enhanced tolerance to infection. These seemingly paradoxical effects suggest that the HRG may serve as a master regulator of systemic homeostasis [11]. However, the mechanisms governing HRG production and secretion remain poorly understood. While current theories attribute sepsis-induced HRG depletion primarily to consumption, this explanation appears insufficient given that other acute-phase proteins with similar consumption patterns actually increase in serum concentration. This discrepancy underscores the need to elucidate HRG regulatory mechanisms, which could provide crucial insights into maintaining homeostasis during infection and potentially lead to clinical applications. Our findings revealed that hepatic HRG reduction parallels a decrease in serum HRG levels during sepsis, suggesting that impaired production rather than secretion defects drive HRG depletion; HRG's N-terminal lysine residue (a destabilizing residue per the N-terminal rule) likely targets it for ubiquitin-mediated degradation [54]. However, there have been no reports on the specific ubiquitination degradation pathway for HRG. Our study revealed that Mid1 is the E3 ubiquitin ligase of HRG and significantly upregulated hepatic Mid1 during sepsis, which explains the decrease in serum HRG levels. These discoveries illuminate a potential physiological regulatory mechanism: early in infection, Mid1-mediated HRG suppression may promote neutrophil activation (HRG normally maintains neutrophil quiescence) [9]. However, during the progression of sepsis, when excessive neutrophil activity causes harmful inflammation and immunosuppression, the body requires HRGs to combat bacterial infection. Unfortunately, ER-stressed hepatocytes may lose their capacity to restore HRG levels through either increased synthesis or decreased degradation. This study demonstrated that inhibition of the E3 ubiquitin ligase Mid1 significantly increases both the intracellular and serum HRG levels in septic mice, leading to improved systemic conditions and increased survival rates. Quantitative analysis revealed that Mid1 inhibition restored the serum HRG concentration in septic mice to levels comparable to those in healthy controls. These findings suggest that targeting hepatocyte Mid1 could serve as a potential therapeutic alternative to exogenous HRG supplementation for the treatment of sepsis.

Small interfering RNA (siRNA) enables sequence-specific gene

silencing and has emerged as a promising therapeutic strategy [55]. The GalNAc-siRNA conjugate platform has successfully achieved clinical translation for liver-targeted delivery due to its efficacy and safety profile [56]. This approach exploits the high specificity of asialoglycoprotein receptor (ASGPR) expression on hepatocytes, for which GalNAc serves as an ideal ligand [57]. In our study, GalNAc-modified siRNA targeting Mid1 effectively suppressed hepatic Mid1 upregulation and ameliorated infection severity in CLP model mice. It accumulates in the kidneys (Fig. S6B) due to abundant blood perfusion, the filtration function of the glomeruli, and other reasons [56]. Pre-treatment (24 h before CLP) is necessary because of delayed cytoplasmic availability post-endocytosis, as siRNA must first traverse endosomal compartments where degradation may occur [58]. These findings suggest the need for further optimization of GalNAc-siMid1 to enhance the targeting precision, metabolic stability, and endosomal escape capability, thereby improving its therapeutic potential in clinical sepsis scenarios.

DPP4 is well-known for its ability to degrade glucagon-like peptide-1 (GLP-1), thereby influencing blood glucose levels. Given this role, DPP4 inhibitors, which specifically target the enzyme activity, have become second-line therapeutics for diabetes. Interestingly, recent studies have revealed an additional function of DPP4 in T cells, in which its knockout suppresses Mid1 expression [26]. Based on these findings, we investigated whether Mid1 inhibition could be leveraged to treat sepsis. In an *in vivo* experiment, we administered oral linagliptin, a clinically approved DPP4 inhibitor, to septic mice. These preliminary findings suggest that linagliptin, which modestly reduced hepatic Mid1 mRNA, confers protective effects in sepsis. However, the precise role of Mid1 downregulation in mediating these benefits requires further investigation. While linagliptin is typically used for diabetes, its protective effects in sepsis appear to be independent of GLP-1-mediated mechanisms. Continuous blood glucose monitoring, including assessments of stress-induced hyperglycemia (during preoperative anesthesia) and sepsis-associated hypoglycemia, revealed no significant differences between the linagliptin-treated and control groups. Linagliptin has been reported to exert antithrombotic effects through the NF-κB pathway, likely by downregulating [59]. This mechanism is particularly relevant because antithrombotic activity, alongside bactericidal effects, is a key feature of HRG, a known therapeutic for sepsis. Our findings thus suggest that linagliptin may combat sepsis through multiple pathways. Since hepatocytes may not be the primary target cells for orally administered linagliptin, we will attempt in future studies to develop hepatocyte-

targeted or specific DPP4 inhibition methods to better elucidate the relationship between hepatocyte DPP4 and Mid1.

4. Conclusions

In summary, we found that Mid1 in hepatocytes may play a crucial role in the pathological process of sepsis. It mediates the infection response of hepatocytes by promoting the degradation of HRG. Based on these findings, we developed GalNAc-siMid1 and linagliptin therapies, which offer new insights into the targeted treatment of sepsis (Fig. 8).

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.lfs.2025.124152>.

CRediT authorship contribution statement

Junjie Shu: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Wei Ma:** Writing – original draft, Methodology, Funding acquisition, Formal analysis. **Haoru Liu:**

Methodology, Investigation, Formal analysis. **Haisu Dai:** Methodology, Investigation. **Hao Zhang:** Methodology, Formal analysis, Data curation. **Rui Gao:** Methodology, Investigation, Formal analysis. **Xue Yang:** Methodology, Investigation, Formal analysis. **Wanqi Tang:** Methodology, Investigation, Formal analysis. **Yang Huang:** Methodology, Formal analysis. **Xiaoyuan Ma:** Methodology, Formal analysis. **Huaping Liang:** Investigation, Formal analysis. **Xia Yang:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization.

Ethics approval and consent for participate

This work has received approval for research ethics from All animal experiments were approved by the Animal Ethics Committee of the Army Medical University (approval no. AMUWEC20247132). The samples from infected liver tissue from patients with intrahepatic bile duct stones complicated by acute infectious cholangitis were approved by the Ethics Committee of the Army Medical Center (approval no. YYLS2025038).

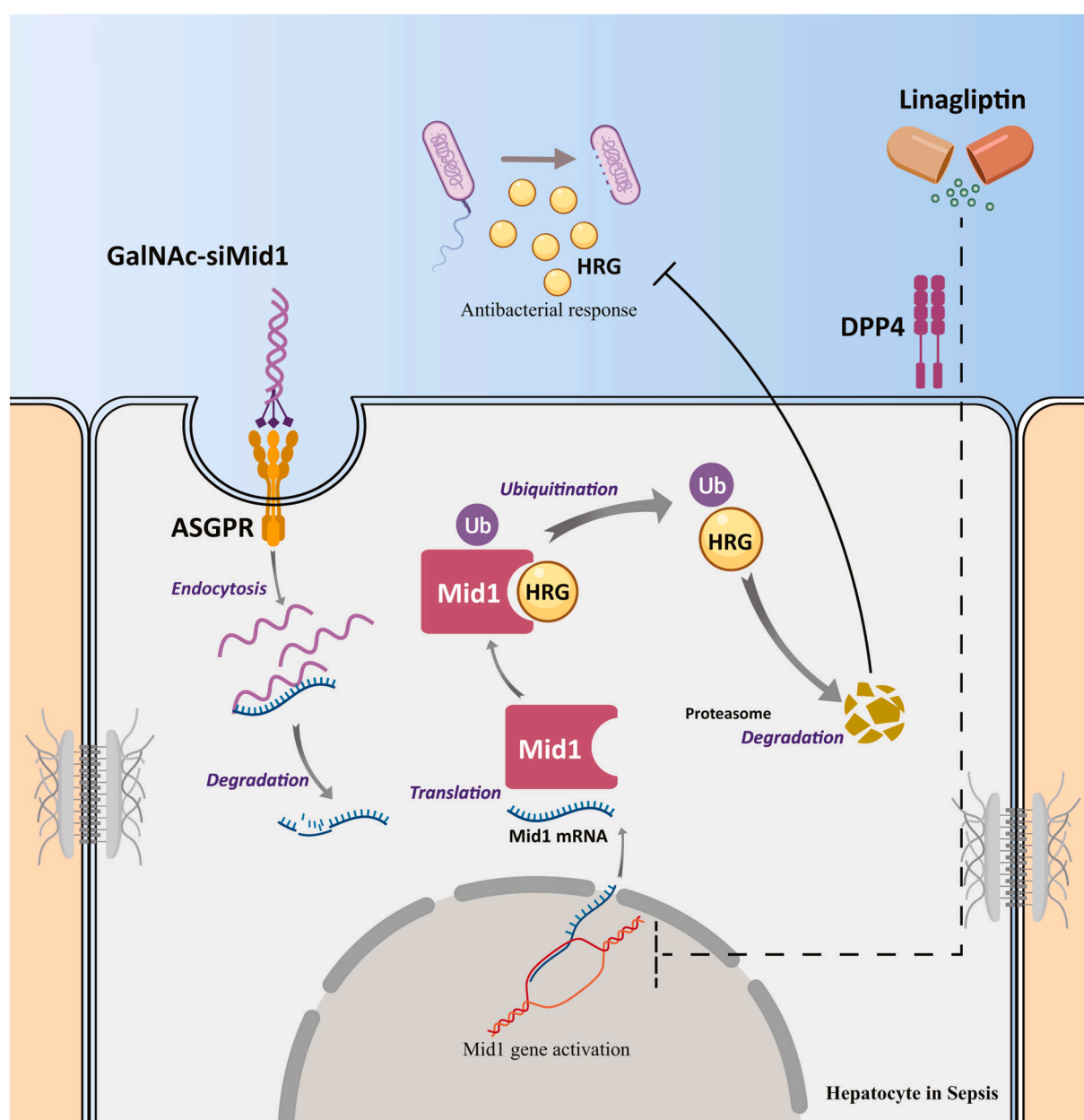


Fig. 8. The Mid1-HRG ubiquitination axis critically regulates hepatocytes contributing to intrinsic antibacterial activity and influencing sepsis progression.

Funding

This study was supported by the National Natural Science Foundation of China (82172134 , 82072227) and some special fund (2023XJS46, 2022-JCJQ-ZD-224-12).

Declaration of competing interest

The authors declare no competing interests.

Acknowledgements

Not applicable.

Data availability

Data will be made available on request.

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